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**UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 2, 2023/2024**

SUBJECT CODE : **SECI1143**

SUBJECT NAME : **PROBABILITY & STATISTICAL DATA ANALYSIS**

SECTION : **01 - 12**

TIME :

DATE/DAY :

VENUES :

INSTRUCTIONS:

- i. Answer **ALL** questions in the given **ANSWER BOOKLET**.
- ii. This paper consists of **SIX (6) QUESTIONS**:
- iii. **DO ALL CALCULATION IN 3 DECIMAL POINTS**.
- iv. Submit **BOTH Questions and Answer Booklet** at the end of the exam.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **EIGHT (8)** printed pages including this page.

STRUCTURED QUESTIONS (100 MARKS)**INSTRUCTION:**

This section consists of **SIX (6)** questions. Answer **ALL** questions in the answer booklet.

QUESTION 1**[10 MARKS]**

- a) A study in agriculture industry is conducted to estimate the average yield of wheat per acre in the farms. The study wants to determine the population mean yield and construct a confidence interval to provide a range within which the true population mean lies. Hence, 15 wheat fields are selected randomly from the farms and record the yields (in KG per acre). The data are as in **Table 1**:

Table 1: Data on yield of wheat per acre in the farms

Wheat fields	1	2	3	4	5
Yields (KG / acre)	50	55	48	52	49
Wheat fields	6	7	8	9	10
Yields (KG / acre)	56	53	47	51	50
Wheat fields	11	12	13	14	15
Yields (KG / acre)	54	52	48	50	55

- i) Calculate a point estimate of the population mean, μ .
(2 Marks)
- ii) Construct and interpret a 90% confidence interval for μ .
(5 Marks)
- b) A financial analyst wants to determine if a sample of stock returns comes from a population with a mean return greater than 8%. The analyst obtains a simple random sample of 60 stock returns from a particular sector. The sample mean return is 9.2%, and the population standard deviation of stock returns in this sector is known to be 3.5%. The analyst decides to test the claim using a significance level of 0.05. Interpret and proof his assumption with statistic.

(3 Marks)

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Wheat fields	11	12	13	14	15
Yields (KG / acre)	54	52	48	50	55

i) Calculate a point estimate of the population mean, μ .

(2 Marks)

ii) Construct and interpret a 90% confidence interval for μ .

(5 Marks)

$$i) \frac{\sum x}{n} = 51.33$$

$$ii) \bar{x} \pm t \left(\frac{s}{\sqrt{n}} \right)$$

$$df = 15 - 1 = 14$$

$$t_{0.05, 14} = 1.761$$

$$51.33 \pm (1.761) \left(\frac{2.82}{\sqrt{15}} \right)$$

...

$$= 50.04 \quad = 52.61$$

$$(50.04, 52.61)$$

I am 90% confident ... between 50.04

b) A financial analyst wants to determine if a sample of stock returns comes from a population with a mean return greater than 8%. The analyst obtains a simple random sample of 60 stock returns from a particular sector. The sample mean return is 9.2%, and the population standard deviation of stock returns in this sector is known to be 3.5%. The analyst decides to test the claim using a significance level of 0.05. Interpret and proof his assumption with statistic.

(3 Marks)

$$b) \mu > 0.08$$

$$\bar{x} = 0.092$$

$$\sigma = 3.5$$

$$Z = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$$\alpha = 0.05$$

$$Z = 1.645$$

$$Z = \frac{0.092 - 0.08}{0.035 / \sqrt{60}}$$

$$= 2.656$$



Reject H_0 , there is sufficient evidence that to support the sample of stock returns greater than 8%.

$$H_0: \mu = 8$$

$$H_1: \mu > 0.08$$

QUESTION 2**[20 MARKS]**

- a) A company is interested in comparing customer satisfaction scores for two different versions of its product. They collected satisfaction ratings from two independent samples of customers who used Version A and Version B of the product. The goal is to determine if there is a significant difference in customer satisfaction between the two versions. The summary of the data is in **Table 2**:

Table 2: Data on customer satisfaction scores for two different versions product

	Number of respondents, n_i	Mean, $\bar{x}_i$	Standard deviation, s_i
Version A	50	4.2	0.8
Version B	60	4.5	0.7

- i) State the hypothesis statement.
(2 Marks)
- ii) Calculate the test statistic.
(2 Marks)
- iii) Calculate the degree of freedom.
(2 Marks)
- iv) Test at $\alpha = 0.05$ level of significance whether there is a mean difference in customer satisfaction between the two versions.
(4 Marks)
- b) A group of students are evaluated to test their performance before and after a tutoring program, and the results are shown in **Table 3**.

Table 3: Data on performance before and after a tutoring program

Before tutoring	72	68	75	80	78
After tutoring	82	85	88	90	86

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- Calculate the test statistic. (2 Marks)
- Calculate the degree of freedom. (2 Marks)
- Test at $\alpha = 0.05$ level of significance whether there is a mean difference in customer satisfaction between the two versions. (4 Marks)

$$i) H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2$$

$$ii) T_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{4.2 - 4.5 - 0}{\sqrt{\frac{0.8^2}{50} + \frac{0.7^2}{60}}}$$

$$= -2.072$$

$$iii) v = \dots \text{long formula one}$$

$$v = 98.25 ?$$

$$v \approx 98$$

$$iv) \alpha = 0.05$$

$$t_{0.05, 98} = -1.96$$



reject H_0 , there is sufficient evidence at the 5% s) to conclude there is a mean difference in the mean customer...

- b) A group of students are evaluated to test their performance before and after a tutoring program, and the results are shown in **Table 3**.

Table 3: Data on performance before and after a tutoring program

Before tutoring	72	68	75	80	78
After tutoring	82	85	88	90	86

$$D: \begin{array}{cccccc} 0 & -10 & -17 & -13 & -10 & -8 \\ D^2 & 100 & 289 & & & \end{array}$$

$$\sum D = -58$$

$$\sum D^2 = 722$$

- State the null and alternative hypotheses (1 Mark)
- By using a significance level of $\alpha = 0.05$, does a tutoring program have a significant effect on the improvement of test scores for students? (9 Marks)

related, match pair

$$i) H_0: \mu_D = 0$$

$$H_1: \mu_D < 0$$

One tail $\alpha = 0.05$

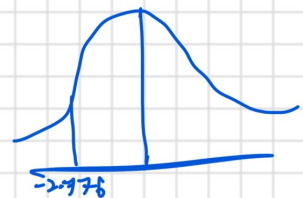
$$t_{0.05}$$

$$ii) t = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}}$$

$$\bar{D} = \frac{\sum D}{n} = \frac{-58}{5} = -11.6$$

$$S_D = \sqrt{\frac{\sum D^2 - \frac{(\sum D)^2}{n}}{n-1}} = \sqrt{\frac{722 - \frac{(-58)^2}{5}}{4}} = 3.507$$

$$t = \frac{-11.6 - 0}{\frac{3.507}{\sqrt{5}}} = -7.396$$



$$\alpha = 0.05$$

$$df = n - 1 = 5 - 1 = 4$$

$$t_{0.05, 4} = 2.132$$

Since $-7.396 < -2.776$ we reject H_0 , there is sufficient evidence

- i) State the null and alternative hypotheses

(1 Mark)

- ii) By using a significance level of $\alpha = 0.05$, does a tutoring program have a significant effect on the improvement of test scores for students?

(9 Marks)

QUESTION 3

[20 MARKS]

- a) A restaurant manager wants to analyse whether the distribution of customer reservations is uniform across weekdays. The manager recorded the number of reservations made each day during a randomly chosen week. The data is presented in **Table 4** below:

Table 4: Data on number of reservations made each day during a randomly chosen week

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Reservation	30	25	35	28	40

- i) Based on this data, conduct a hypothesis test.
- (1 Mark)
- ii) Calculate the expected frequency and the test statistic.
- (4 Marks)
- iii) Find the critical value. Do the data prove that the number of reservations is equally distributed throughout the weekdays. Use significance level, $\alpha = 0.05$.
- (3 Marks)
- b) Internet has become the most important means of communication and gathering information. People with different age use Internet for different purposes. A random sample of 159 adults yielded the data shows in **Table 5** to find out the relationship between age and the usage of internet.

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$$E = \frac{n}{k}$$

- i) Based on this data, conduct a hypothesis test. (1 Mark)
- ii) Calculate the expected frequency and the test statistic. (4 Marks)
- iii) Find the critical value. Do the data prove that the number of reservations is equally distributed throughout the weekdays. Use significance level, $\alpha = 0.05$. (3 Marks)

$$H_0: p_1 = p_2 = p_3 = p_4 = p_5$$

H_a : at least one of the probability is different from others:

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Reservation	30	25	35	28	40

$$E = \frac{n}{k} = \frac{158}{5} = 31.6$$

$$\frac{(O-E)^2}{E}$$

0.081	1.378	0.366	0.410	2.233
-------	-------	-------	-------	-------

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 4.468$$

$$iii) d = 0.05$$

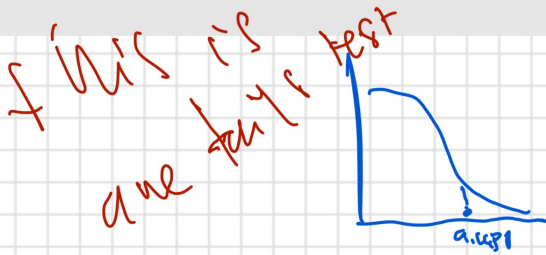
$$df = k - 1$$

$$= 5 - 1$$

$$= 4$$

$$\chi^2_{0.05, 4} = 9.488$$

Since $4.468 < 9.488$, we do not reject H_0 .



- b) Internet has become the most important means of communication and gathering information. People with different age use Internet for different purposes. A random sample of 159 adults yielded the data shows in **Table 5** to find out the relationship between age and the usage of internet.

Table 5: The relationship between age and the usage of internet

Internet Usage		Age	
		25- 64	65+
Internet Usage	Never	38	31
	Sometimes	31	4
	Every day	50	5

- i) State the null and alternative hypotheses. (2 Marks)
- ii) Calculate the expected count and the test statistic (7 Marks)
- iii) At the 1% significance level, do the data provide sufficient evidence to conclude that a relationship exists between age and Internet usage? (3 Marks)

$$i) H_0: \text{Age and internet usage are independent}$$

$$H_a: \text{Age and internet usage are dependent}$$

ii)

Internet Usage		Age	
		25- 64	65+
Internet Usage	Never	38	31
	Sometimes	31	4
	Every day	50	5
		119	40

	O	E	$\frac{(O-E)^2}{E}$
N(25-64)	38	51.64	3.60
N(65+)	31	17.36	10.72
S(25-64)	31	20.19	0.88
S(65+)	4	8.81	2.62
E(25-64)	50	41.16	1.90
E(65+)	5	13.84	5.66

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 25.38$$

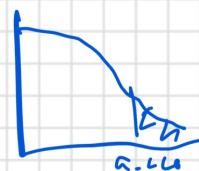
$$iii) d = 0.01$$

$$df = (r-1)(c-1)$$

$$= (3-1)(2-1)$$

$$= 2$$

$$\chi^2_{0.01, 2} = 9.210$$



reject H_0 , since...

Table 5: The relationship between age and the usage of internet

		Age	
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Internet Usage	Never	38	31
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- i) State the null and alternative hypotheses. (2 Marks)
- ii) Calculate the expected count and the test statistic (7 Marks)
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QUESTION 4**[15 MARKS]**

Table 6 shows the blood pressure, p mmHG, and albumin, a , mg/gram level for 7 patients. High blood pressure may cause kidney disease or damage indicated by albumin level. An analysis must be conducted to understand the relationship between blood pressure and raised albumin levels.

Table 6: Blood pressure and albumin level

Patient	A	B	C	D	E	F	G
p	160	162	175	185	178	170	182
a	22	29	31	35	33	38	45

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Patient	A	B	C	D	E	F	G
p	160	162	175	185	178	170	182
a	22	29	31	35	33	38	45

should be x
y

- a) Draw (sketch) the scatterplot. Is there any relationship observed? If yes, determine whether the relationship is negative or positive.

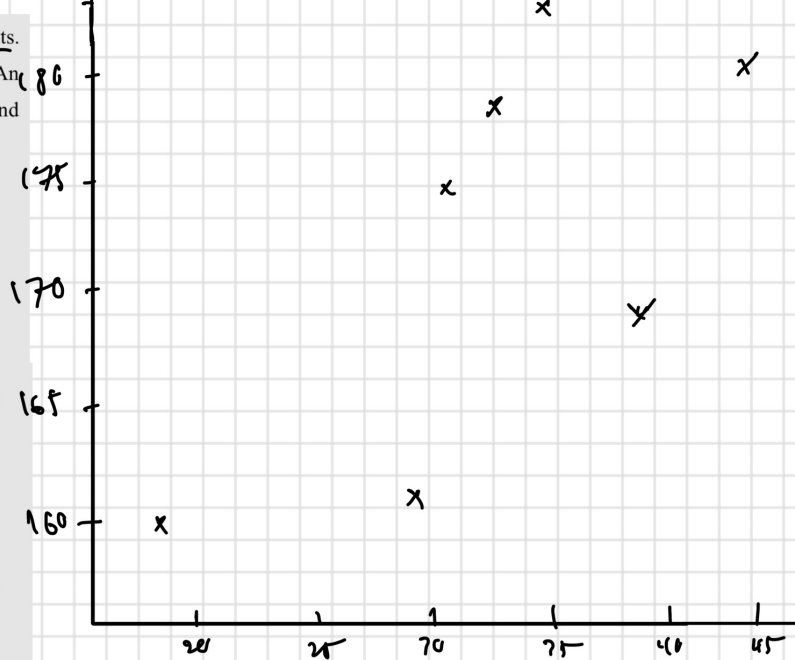
(2 Marks)

- b) Calculate the correlation coefficient for data in **Table 6**, given $\Sigma p^2 = 210402$ and $\Sigma a^2 = 8069$. Interpret the result.

(11 Marks)

- c) Each patient in **Table 6** is given medication that lowers the blood pressure level, p , what will happen to albumin level, a , based on your findings in (b)?

(2 Marks)



There is relationship, as albumin increase, blood pressure increase

b)

Patient	A	B	C	D	E	F	G
p	160	162	175	185	178	170	182
a	22	29	31	35	33	38	45

$$\Sigma p = 1212 \quad \Sigma p^2 = 210402$$

$$\Sigma a = 233 \quad \Sigma a^2 = 8069$$

$$\Sigma xy = 40642$$

$$r = \frac{40642 - \frac{(1212)(233)}{7}}{\sqrt{\left(210402 - \frac{(1212)^2}{7}\right)\left(8069 - \frac{(233)^2}{7}\right)}}$$

$$= 0.8$$

The correlation coefficient is approximately 0.800, indicating a strong positive linear relationship between blood pressure and albumin level.

close to 1

c) albumin level is expected to decrease as well

- a) Draw (sketch) the scatterplot. Is there any relationship observed? If yes, determine whether the relationship is negative or positive.

(2 Marks)

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QUESTION 5

[15 MARKS]

- a) Kembara Motor Association (KMA) is analyzing the relationship between engine volume and mileage, given the same volume of petrol. **Table 7** is tabulated for liquid-cooled 4-stroke 16-valve cylinder engines from different motorbikes manufacturers. The engine volume is in total cubic km.

Table 7: Engine Volume and Mileage of different Motorbikes Manufacturer

Motorbikes manufacturer	Engine volume, x (cubic km)	Mileage, y (km)
A	97	24
B	88	21
C	105	24
D	151	25
E	140	22
F	134	23
G	118	22

Based on the given data,

- i) Find the summation for x , y , x^2 , y^2 and xy for the above data.

(5 Marks)

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G	118	22

Based on the given data,

i) Find the summation for x, y, x², y² and xy for the above data.

(5 Marks)

ii) Find the linear regression equation using the least-square method.

(4 Marks)

iii) Use the linear regression equation to estimate the mileage for a motorbike which has 120 cubic km engine volume. x

(1 Mark)

$$\begin{aligned} \text{iii)} \quad \hat{y} &= 20.501 + 0.021(120) \\ &= 23.021 \end{aligned}$$

b) Dr. Fatimah is monitoring 6 diabetic patients in her ward. After her morning round, she observed that her patient' blood pressure, (p mmHg) and their age (t years) as in Table 8 below:

Table 8: Blood pressure ages of Dr. Fatimah's patients

Patient	1	2	3	4	5	6
t	35	74	48	60	42	26
p	88	130	120	135	98	80

i) Sketch a scatter diagram of the data and interpret the correlation coefficient.

(2 marks)

ii) Given the estimator of the standard error of the slope (s_{b_1}) is 0.07 and the linear regression equation for the data is $\hat{y} = 50.728 + 1.216x$, does the patient's age affect the blood pressure reading? Test the hypothesis using 95% confidence level.

(3 marks)

$$\begin{aligned} s_{b_1} &= 0.07 \\ \hat{y} &= 50.728 + 1.216x \\ b_0 &= 50.728 \\ b_1 &= 1.216 \end{aligned}$$

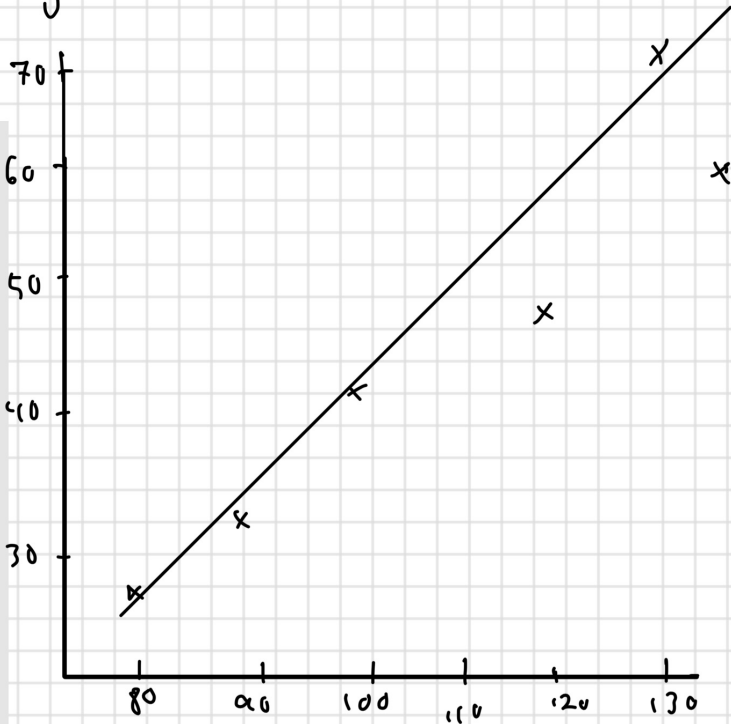
$$\begin{aligned} H_0 &: \beta = 0 \\ H_1 &: \beta \neq 0 \\ t &= \frac{b_1 - \beta_1}{s_{b_1}} \\ &= \frac{1.216 - 0}{0.07} \\ &= 17.37 \end{aligned}$$

$$\begin{aligned} \alpha &= 0.05 \\ df &= n - 2 \\ &= 6 - 2 \\ &= 4 \\ t_{0.025, 4} &= 2.776 \end{aligned}$$

$$17.37 > 2.776, \text{ reject } H_0, \dots$$

$$\begin{aligned} \text{i)} \quad \sum x &= 833 \\ \sum y &= 161 \\ \sum x^2 &= 102459 \\ \sum y^2 &= 3715 \\ \sum xy &= 19229 \\ \text{ii)} \quad y &= b_0 + b_1x \\ b_1 &= \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} \quad b_0 = \bar{y} - b_1\bar{x} \\ b_1 &= \frac{19229 - \frac{(833)(161)}{7}}{102459 - \frac{(833)^2}{7}} \\ &= 0.021 \\ b_0 &= (23) - 0.021(119) \\ &= 20.501 \end{aligned}$$

$$\hat{y} = 20.501 + 0.021x$$



- ii) Find the linear regression equation using the least-square method.
(4 Marks)
- iii) Use the linear regression equation to estimate the mileage for a motorbike which has 120 cubic km engine volume.
(1 Mark)
- b) Dr. Fatimah is monitoring 6 diabetic patients in her ward. After her morning round, she observed that her patient's blood pressure, (p mmHg) and their age (t years) as in **Table 8** below:

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(3 marks)

QUESTION 6**[20 MARKS]**

Table 9 below examines departure delay times for AM Airline flights from Senai Johor to Kuching Sarawak. Departure delay times, measured in minutes, reflect the time difference between scheduled departure and actual departure.

The negative values denote flights departing earlier than scheduled (early take-off) and 0 means the flight departed on time (no delay). The objective of the study is to ascertain whether these departure delay times vary significantly across different flights.

Table 9: Departure delay times for AM Airline flights

Flight No	Departure Delay Times (minutes)								Standard Deviation, s_i
AB103	-2	-1	-2	2	-2	0	-2	-3	$s_1 = 1.58$
AB112	19	-4	-5	-1	-4	73	0	1	$s_2 = 26.64$
AB108	18	60	142	-1	-11	-1	47	13	$s_3 = 50.25$

Let: $\bar{x}_1$ is the mean for flight AB103.
 $\bar{x}_2$ is the mean for flight AB112.
 $\bar{x}_3$ is the mean for flight AB108.

a) Find:

i) $\bar{x}$ for each category.

(3 Marks)

ii) Variance between samples.

(5 Marks)

iii) Variance within samples.

(4 Marks)

b) Use a 0.05 significance level to test the claim that the different flights have the same mean departure delay time.

(8 Marks)

QUESTION 6

[20 MARKS]

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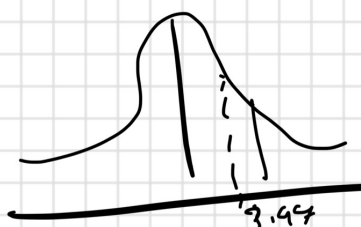
$$b) F = \frac{MS_{\text{B}}}{MS_{\text{P}}} = \frac{2500.66}{1079.08}$$

$$F = 2.317$$

$$df_{\text{num}} = k - 1 = 3 - 1 = 2$$

$$df_{\text{denom}} = k(h - 1) = 3(8 - 1) = 21$$

$$F_{(2, 21, 0.05)} = 3.47$$



failed to reject H_0 .

$$a) i) \bar{x}_1 = -1.25$$

$$\bar{x}_2 = 9.875$$

$$\bar{x}_3 = 33.375$$

$$ii) \bar{\bar{x}} = \frac{-1.25 + 9.875 + 33.375}{3} = 14$$

$$S_x = \sqrt{\frac{(-1.25 - 14)^2 + (9.875 - 14)^2 + (33.375 - 14)^2}{3 - 1}}$$

$$= 17.68$$

$$MS_{\text{B}} = \frac{8(17.68)^2}{2500.66}$$

$$iii) \frac{(1.58)^2 + (26.64)^2 + (50.25)^2}{3}$$

$$= 1079.08$$

-END OF QUESTIONS PAPER-**List of Formula:**

(a) Hypothesis Testing for one sample:

p value $\rightarrow p \leq \alpha \rightarrow$ reject
 $p > \alpha \rightarrow$ fail to reject
 2 tail $= p \times 2$

H_0 : value
 Test Statistic, $z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$

test on mean
 variance unknown
 $df = n - 1$
 Test Statistic, $t_0 = \frac{\bar{x} - \mu}{s / \sqrt{n}}$

test on mean
 variance known
 H_0 : value
 Test Statistic, $z_0 = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

test on variance
 $df = n - 1$, use right tail $= 1 - \alpha$ (if same)
 Test Statistic, $\chi^2 = \frac{(n - 1)s^2}{\sigma^2}$

Sample proportion:

$$\hat{p} = \frac{\text{Selected}}{n, \text{total}}$$

margin error SE

$$\hat{p} \pm (z) \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = (x, y)$$

Calculate from data

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

sample mean
= SX in calc σ is unknown

take the 2 tail t

$$\bar{x} \pm (t) \left(\frac{s}{\sqrt{n}} \right)$$

$df = n - 1$

(b) Hypothesis Testing for two samples:

pooled sample proportion

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

$$\bar{q} = 1 - \bar{p}$$

$$p_1 - p_2 = 0$$

$H_0: p_1 = p_2$
 $H_1: p_1 \neq p_2$
 Test Statistic, $z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$

test on mean
 variance known
 $\mu_1 - \mu_2 = \Delta_0$
 $\mu_1 - \mu_2 = 0$
 Test Statistic, $z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

test on mean
 variance unknown
 $\sigma_1 = \sigma_2$ equal variance
 $\mu_1 - \mu_2 = 0$
 $\mu_1 \neq \mu_2$
 Test Statistic, $t_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$; $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
 $df = n_1 + n_2 - 2$

test on mean
 variance unknown
 $\sigma_1 \neq \sigma_2$ unequal variance
 Test Statistic, $T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$; $v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$
 degrees of freedom
 not drunk!!
 round down to lowest
 ≈ 13.2
 ≈ 13
 $- 9 -$

difference between variance/SD

$$\text{Test Statistic, } F_0 = \frac{s_1^2}{s_2^2}; \text{ Degree of Freedom} = F_{(n_1-1, n_2-1)}$$

test on matched pair

$$\text{Test Statistic, } t = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}}; \text{ Degree of Freedom} = n - 1$$

$\bar{D} = \frac{\sum D}{n}$; $S_D = \sqrt{\frac{\sum D^2 - \frac{(\sum D)^2}{n}}{n-1}}$

(c) Chi-Square Test

$$\text{Test Statistic, } \chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i} \quad E = \frac{n}{k} \quad E = np$$

Degree of Freedom for one-way contingency table = $k - 1$

$$\text{Test Statistic, } \chi^2 = \sum_i \frac{(o_{ij} - e_{ij})^2}{e_{ij}}, \quad e_{ij} = \frac{(i^{\text{th}} \text{ Row total})(j^{\text{th}} \text{ Column total})}{\text{Total sample size}}$$

Degree of Freedom for two-way contingency table = $(r - 1)(c - 1)$

(d) Correlation

Pearson Correlation Coefficient,

$$r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

Spearman Correlation Coefficient,

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \text{ where } d_i = y_i - x_i$$

$$\text{Test Statistic, } t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

$H_0: \rho = 0$ (no linear correlation) $df = n-2$
 $H_1: \rho \neq 0$ (linear correlation exist) $\rightarrow t$ table

(e) Regression

$$H_0: \beta = 0$$

$$H_1: \beta \neq 0$$

$$b_1 = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} ; b_0 = \bar{y} - b_1 \bar{x}$$

$$SST = SSE + SSR$$

$$R^2 = \frac{SSR}{SST}$$

$$Test\ Statistic, t = \frac{b_1 - \beta_1}{S_{b_1}} ; Degree\ of\ Freedom = n - 2$$

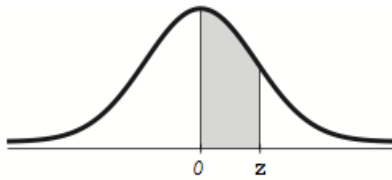
(f) Analysis of variance (ANOVA)

$$Test\ Statistic, F = \frac{ns_{\bar{x}}^2}{s_p^2}$$

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_1: \text{at least 1 mean is different}$$

- numerator degrees of freedom = $k - 1$
- denominator degrees of freedom = $k(n - 1)$

**TABLE A-2** Standard Normal (z) Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	*.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	↑.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	*.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	↑.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.10 and higher	.4999									

NOTE: For values of z above 3.09, use 0.4999 for the area.

*Use these common values that result from interpolation:

z score	Area
1.645	0.4500 ←
2.575	0.4950 ←

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STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

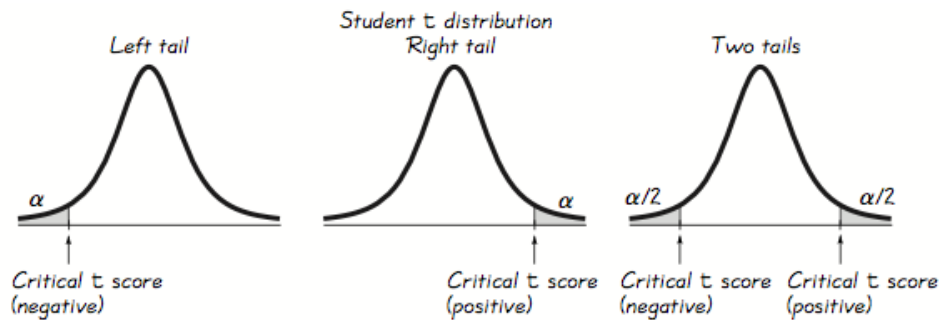
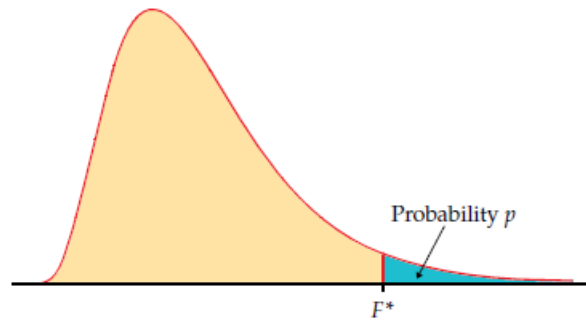


TABLE A-3 t Distribution						
α						
Degrees of Freedom	.005 (one tail) .01 (two tails)	.01 (one tail) .02 (two tails)	.025 (one tail) .05 (two tails)	.05 (one tail) .10 (two tails)	.10 (one tail) .20 (two tails)	.25 (one tail) .50 (two tails)
1	63.657	31.821	12.706	6.314	3.078	1.000
2	9.925	6.965	4.303	2.920	1.886	.816
3	5.841	4.541	3.182	2.353	1.638	.765
4	4.604	3.747	2.776	2.132	1.533	.741
5	4.032	3.365	2.571	2.015	1.476	.727
6	3.707	3.143	2.447	1.943	1.440	.718
7	3.500	2.998	2.365	1.895	1.415	.711
8	3.355	2.896	2.306	1.860	1.397	.706
9	3.250	2.821	2.262	1.833	1.383	.703
10	3.169	2.764	2.228	1.812	1.372	.700
11	3.106	2.718	2.201	1.796	1.363	.697
12	3.054	2.681	2.179	1.782	1.356	.696
13	3.012	2.650	2.160	1.771	1.350	.694
14	2.977	2.625	2.145	1.761	1.345	.692
15	2.947	2.602	2.132	1.753	1.341	.691
16	2.921	2.584	2.120	1.746	1.337	.690
17	2.898	2.567	2.110	1.740	1.333	.689
18	2.878	2.552	2.101	1.734	1.330	.688
19	2.861	2.540	2.093	1.729	1.328	.688
20	2.845	2.528	2.086	1.725	1.325	.687
21	2.831	2.518	2.080	1.721	1.323	.686
22	2.819	2.508	2.074	1.717	1.321	.686
23	2.807	2.500	2.069	1.714	1.320	.685
24	2.797	2.492	2.064	1.711	1.318	.685
25	2.787	2.485	2.060	1.708	1.316	.684
26	2.779	2.479	2.056	1.706	1.315	.684
27	2.771	2.473	2.052	1.703	1.314	.684
28	2.763	2.467	2.048	1.701	1.313	.683
29	2.756	2.462	2.045	1.699	1.311	.683
Large (z)	2.575	2.326	1.960	1.645	1.282	.675

TABLE A-4		Chi-Square (χ^2) Distribution								
Degrees of Freedom	Area to the Right of the Critical Value									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

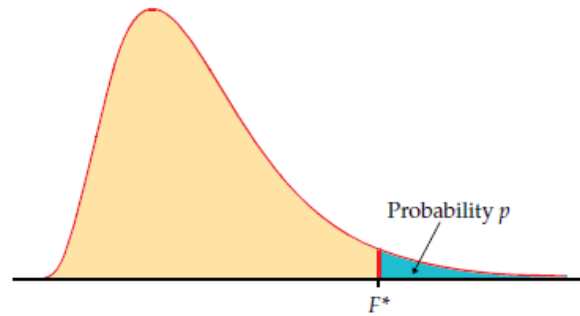
T-12 Tables

Table entry for p is the critical value F^* with probability p lying to its right.

**TABLE E***F* critical values

		Degrees of freedom in the numerator								
	p	1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86
	.050	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
	.025	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28
	.010	4052.2	4999.5	5403.4	5624.6	5763.6	5859.0	5928.4	5981.1	6022.5
	.001	405284	500000	540379	562500	576405	585937	592873	598144	602284
	.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38
	.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
	.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39
	.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
	.001	998.50	999.00	999.17	999.25	999.30	999.33	999.36	999.37	999.39
	.100	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24
	.050	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
	.025	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47
	.010	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35
	.001	167.03	148.50	141.11	137.10	134.58	132.85	131.58	130.62	129.86
	.100	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94
	.050	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
	.025	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90
	.010	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
	.001	74.14	61.25	56.18	53.44	51.71	50.53	49.66	49.00	48.47
	.100	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32
	.050	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
	.025	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68
	.010	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16
	.001	47.18	37.12	33.20	31.09	29.75	28.83	28.16	27.65	27.24
	.100	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96
	.050	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
	.025	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52
	.010	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98
	.001	35.51	27.00	23.70	21.92	20.80	20.03	19.46	19.03	18.69
	.100	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72
	.050	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
	.025	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82
	.010	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
	.001	29.25	21.69	18.77	17.20	16.21	15.52	15.02	14.63	14.33

Table entry for p is the critical value F^* with probability p lying to its right.

**TABLE E***F* critical values (continued)

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
60.19	60.71	61.22	61.74	62.05	62.26	62.53	62.69	62.79	63.06	63.30
241.88	243.91	245.95	248.01	249.26	250.10	251.14	251.77	252.20	253.25	254.19
968.63	976.71	984.87	993.10	998.08	1001.4	1005.6	1008.1	1009.8	1014.0	1017.7
6055.8	6106.3	6157.3	6208.7	6239.8	6260.6	6286.8	6302.5	6313.0	6339.4	6362.7
605621	610668	615764	620908	624017	626099	628712	630285	631337	633972	636301
9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.47	9.48	9.49
19.40	19.41	19.43	19.45	19.46	19.46	19.47	19.48	19.48	19.49	19.49
39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.48	39.49	39.50
99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.48	99.49	99.50
999.40	999.42	999.43	999.45	999.46	999.47	999.47	999.48	999.48	999.49	999.50
5.23	5.22	5.20	5.18	5.17	5.17	5.16	5.15	5.15	5.14	5.13
8.79	8.74	8.70	8.66	8.63	8.62	8.59	8.58	8.57	8.55	8.53
14.42	14.34	14.25	14.17	14.12	14.08	14.04	14.01	13.99	13.95	13.91
27.23	27.05	26.87	26.69	26.58	26.50	26.41	26.35	26.32	26.22	26.14
129.25	128.32	127.37	126.42	125.84	125.45	124.96	124.66	124.47	123.97	123.53
3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.80	3.79	3.78	3.76
5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.70	5.69	5.66	5.63
8.84	8.75	8.66	8.56	8.50	8.46	8.41	8.38	8.36	8.31	8.26
14.55	14.37	14.20	14.02	13.91	13.84	13.75	13.69	13.65	13.56	13.47
48.05	47.41	46.76	46.10	45.70	45.43	45.09	44.88	44.75	44.40	44.09
3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.15	3.14	3.12	3.11
4.74	4.68	4.62	4.56	4.52	4.50	4.46	4.44	4.43	4.40	4.37
6.62	6.52	6.43	6.33	6.27	6.23	6.18	6.14	6.12	6.07	6.02
10.05	9.89	9.72	9.55	9.45	9.38	9.29	9.24	9.20	9.11	9.03
26.92	26.42	25.91	25.39	25.08	24.87	24.60	24.44	24.33	24.06	23.82
2.94	2.90	2.87	2.84	2.81	2.80	2.78	2.77	2.76	2.74	2.72
4.06	4.00	3.94	3.87	3.83	3.81	3.77	3.75	3.74	3.70	3.67
5.46	5.37	5.27	5.17	5.11	5.07	5.01	4.98	4.96	4.90	4.86
7.87	7.72	7.56	7.40	7.30	7.23	7.14	7.09	7.06	6.97	6.89
18.41	17.99	17.56	17.12	16.85	16.67	16.44	16.31	16.21	15.98	15.77
2.70	2.67	2.63	2.59	2.57	2.56	2.54	2.52	2.51	2.49	2.47
3.64	3.57	3.51	3.44	3.40	3.38	3.34	3.32	3.30	3.27	3.23
4.76	4.67	4.57	4.47	4.40	4.36	4.31	4.28	4.25	4.20	4.15
6.62	6.47	6.31	6.16	6.06	5.99	5.91	5.86	5.82	5.74	5.66
14.08	13.71	13.32	12.93	12.69	12.53	12.33	12.20	12.12	11.91	11.72

(Continued)

T-14 Tables

TABLE E*F* critical values (continued)

		Degrees of freedom in the numerator								
<i>p</i>		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	8	.100	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59
		.050	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44
		.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43
		.010	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03
		.001	25.41	18.49	15.83	14.39	13.48	12.86	12.40	12.05
	9	.100	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47
		.050	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23
		.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10
		.010	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47
		.001	22.86	16.39	13.90	12.56	11.71	11.13	10.70	10.37
	10	.100	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38
		.050	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07
		.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85
		.010	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06
		.001	21.04	14.91	12.55	11.28	10.48	9.93	9.52	9.20
	11	.100	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30
		.050	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95
		.025	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66
		.010	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74
		.001	19.69	13.81	11.56	10.35	9.58	9.05	8.66	8.35
	12	.100	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24
		.050	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85
		.025	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51
		.010	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50
		.001	18.64	12.97	10.80	9.63	8.89	8.38	8.00	7.71
	13	.100	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20
		.050	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77
		.025	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39
		.010	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30
		.001	17.82	12.31	10.21	9.07	8.35	7.86	7.49	7.21
	14	.100	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15
		.050	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70
		.025	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29
		.010	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14
		.001	17.14	11.78	9.73	8.62	7.92	7.44	7.08	6.80
	15	.100	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12
		.050	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64
		.025	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20
		.010	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00
		.001	16.59	11.34	9.34	8.25	7.57	7.09	6.74	6.47
	16	.100	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09
		.050	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59
		.025	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12
		.010	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89
		.001	16.12	10.97	9.01	7.94	7.27	6.80	6.46	6.19
	17	.100	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06
		.050	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55
		.025	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06
		.010	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.79
		.001	15.72	10.66	8.73	7.68	7.02	6.56	6.22	5.96

TABLE E*F* critical values (continued)

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.35	2.34	2.32	2.30
3.35	3.28	3.22	3.15	3.11	3.08	3.04	3.02	3.01	2.97	2.93
4.30	4.20	4.10	4.00	3.94	3.89	3.84	3.81	3.78	3.73	3.68
5.81	5.67	5.52	5.36	5.26	5.20	5.12	5.07	5.03	4.95	4.87
11.54	11.19	10.84	10.48	10.26	10.11	9.92	9.80	9.73	9.53	9.36
2.42	2.38	2.34	2.30	2.27	2.25	2.23	2.22	2.21	2.18	2.16
3.14	3.07	3.01	2.94	2.89	2.86	2.83	2.80	2.79	2.75	2.71
3.96	3.87	3.77	3.67	3.60	3.56	3.51	3.47	3.45	3.39	3.34
5.26	5.11	4.96	4.81	4.71	4.65	4.57	4.52	4.48	4.40	4.32
9.89	9.57	9.24	8.90	8.69	8.55	8.37	8.26	8.19	8.00	7.84
2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11	2.08	2.06
2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62	2.58	2.54
3.72	3.62	3.52	3.42	3.35	3.31	3.26	3.22	3.20	3.14	3.09
4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08	4.00	3.92
8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12	6.94	6.78
2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03	2.00	1.98
2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49	2.45	2.41
3.53	3.43	3.33	3.23	3.16	3.12	3.06	3.03	3.00	2.94	2.89
4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78	3.69	3.61
7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35	6.18	6.02
2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96	1.93	1.91
2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38	2.34	2.30
3.37	3.28	3.18	3.07	3.01	2.96	2.91	2.87	2.85	2.79	2.73
4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54	3.45	3.37
7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76	5.59	5.44
2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90	1.88	1.85
2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30	2.25	2.21
3.25	3.15	3.05	2.95	2.88	2.84	2.78	2.74	2.72	2.66	2.60
4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34	3.25	3.18
6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30	5.14	4.99
2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86	1.83	1.80
2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22	2.18	2.14
3.15	3.05	2.95	2.84	2.78	2.73	2.67	2.64	2.61	2.55	2.50
3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18	3.09	3.02
6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94	4.77	4.62
2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82	1.79	1.76
2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16	2.11	2.07
3.06	2.96	2.86	2.76	2.69	2.64	2.59	2.55	2.52	2.46	2.40
3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05	2.96	2.88
6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64	4.47	4.33
2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78	1.75	1.72
2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11	2.06	2.02
2.99	2.89	2.79	2.68	2.61	2.57	2.51	2.47	2.45	2.38	2.32
3.69	3.55	3.41	3.26	3.16	3.10	3.02	2.97	2.93	2.84	2.76
5.81	5.55	5.27	4.99	4.82	4.70	4.54	4.45	4.39	4.23	4.08
2.00	1.96	1.91	1.86	1.83	1.81	1.78	1.76	1.75	1.72	1.69
2.45	2.38	2.31	2.23	2.18	2.15	2.10	2.08	2.06	2.01	1.97
2.92	2.82	2.72	2.62	2.55	2.50	2.44	2.41	2.38	2.32	2.26
3.59	3.46	3.31	3.16	3.07	3.00	2.92	2.87	2.83	2.75	2.66
5.58	5.32	5.05	4.78	4.60	4.48	4.33	4.24	4.18	4.02	3.87

(Continued)

T-16 Tables

TABLE E*F* critical values (continued)

		Degrees of freedom in the numerator								
<i>p</i>		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	18	.100	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.00
		.050	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.46
		.025	5.98	4.56	3.95	3.61	3.38	3.22	3.10	2.93
		.010	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.60
		.001	15.38	10.39	8.49	7.46	6.81	6.35	6.02	5.56
	19	.100	2.99	2.61	2.40	2.27	2.18	2.11	2.06	1.98
		.050	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.42
		.025	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.88
		.010	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.52
		.001	15.08	10.16	8.28	7.27	6.62	6.18	5.85	5.39
	20	.100	2.97	2.59	2.38	2.25	2.16	2.09	2.04	1.96
		.050	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.39
		.025	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.84
		.010	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.46
		.001	14.82	9.95	8.10	7.10	6.46	6.02	5.69	5.24
	21	.100	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.95
		.050	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.37
		.025	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.80
		.010	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.40
		.001	14.59	9.77	7.94	6.95	6.32	5.88	5.56	5.11
	22	.100	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.93
		.050	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.34
		.025	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.76
		.010	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.35
		.001	14.38	9.61	7.80	6.81	6.19	5.76	5.44	4.99
	23	.100	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.92
		.050	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.32
		.025	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.73
		.010	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.30
		.001	14.20	9.47	7.67	6.70	6.08	5.65	5.33	4.89
	24	.100	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.91
		.050	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.30
		.025	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.70
		.010	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.26
		.001	14.03	9.34	7.55	6.59	5.98	5.55	5.23	4.80
	25	.100	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.89
		.050	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.28
		.025	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.68
		.010	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.22
		.001	13.88	9.22	7.45	6.49	5.89	5.46	5.15	4.71
	26	.100	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.88
		.050	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.27
		.025	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.65
		.010	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.18
		.001	13.74	9.12	7.36	6.41	5.80	5.38	5.07	4.64
	27	.100	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.87
		.050	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.25
		.025	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.63
		.010	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.15
		.001	13.61	9.02	7.27	6.33	5.73	5.31	5.00	4.57

TABLE E*F* critical values (continued)

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
1.98	1.93	1.89	1.84	1.80	1.78	1.75	1.74	1.72	1.69	1.66
2.41	2.34	2.27	2.19	2.14	2.11	2.06	2.04	2.02	1.97	1.92
2.87	2.77	2.67	2.56	2.49	2.44	2.38	2.35	2.32	2.26	2.20
3.51	3.37	3.23	3.08	2.98	2.92	2.84	2.78	2.75	2.66	2.58
5.39	5.13	4.87	4.59	4.42	4.30	4.15	4.06	4.00	3.84	3.69
1.96	1.91	1.86	1.81	1.78	1.76	1.73	1.71	1.70	1.67	1.64
2.38	2.31	2.23	2.16	2.11	2.07	2.03	2.00	1.98	1.93	1.88
2.82	2.72	2.62	2.51	2.44	2.39	2.33	2.30	2.27	2.20	2.14
3.43	3.30	3.15	3.00	2.91	2.84	2.76	2.71	2.67	2.58	2.50
5.22	4.97	4.70	4.43	4.26	4.14	3.99	3.90	3.84	3.68	3.53
1.94	1.89	1.84	1.79	1.76	1.74	1.71	1.69	1.68	1.64	1.61
2.35	2.28	2.20	2.12	2.07	2.04	1.99	1.97	1.95	1.90	1.85
2.77	2.68	2.57	2.46	2.40	2.35	2.29	2.25	2.22	2.16	2.09
3.37	3.23	3.09	2.94	2.84	2.78	2.69	2.64	2.61	2.52	2.43
5.08	4.82	4.56	4.29	4.12	4.00	3.86	3.77	3.70	3.54	3.40
1.92	1.87	1.83	1.78	1.74	1.72	1.69	1.67	1.66	1.62	1.59
2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.94	1.92	1.87	1.82
2.73	2.64	2.53	2.42	2.36	2.31	2.25	2.21	2.18	2.11	2.05
3.31	3.17	3.03	2.88	2.79	2.72	2.64	2.58	2.55	2.46	2.37
4.95	4.70	4.44	4.17	4.00	3.88	3.74	3.64	3.58	3.42	3.28
1.90	1.86	1.81	1.76	1.73	1.70	1.67	1.65	1.64	1.60	1.57
2.30	2.23	2.15	2.07	2.02	1.98	1.94	1.91	1.89	1.84	1.79
2.70	2.60	2.50	2.39	2.32	2.27	2.21	2.17	2.14	2.08	2.01
3.26	3.12	2.98	2.83	2.73	2.67	2.58	2.53	2.50	2.40	2.32
4.83	4.58	4.33	4.06	3.89	3.78	3.63	3.54	3.48	3.32	3.17
1.89	1.84	1.80	1.74	1.71	1.69	1.66	1.64	1.62	1.59	1.55
2.27	2.20	2.13	2.05	2.00	1.96	1.91	1.88	1.86	1.81	1.76
2.67	2.57	2.47	2.36	2.29	2.24	2.18	2.14	2.11	2.04	1.98
3.21	3.07	2.93	2.78	2.69	2.62	2.54	2.48	2.45	2.35	2.27
4.73	4.48	4.23	3.96	3.79	3.68	3.53	3.44	3.38	3.22	3.08
1.88	1.83	1.78	1.73	1.70	1.67	1.64	1.62	1.61	1.57	1.54
2.25	2.18	2.11	2.03	1.97	1.94	1.89	1.86	1.84	1.79	1.74
2.64	2.54	2.44	2.33	2.26	2.21	2.15	2.11	2.08	2.01	1.94
3.17	3.03	2.89	2.74	2.64	2.58	2.49	2.44	2.40	2.31	2.22
4.64	4.39	4.14	3.87	3.71	3.59	3.45	3.36	3.29	3.14	2.99
1.87	1.82	1.77	1.72	1.68	1.66	1.63	1.61	1.59	1.56	1.52
2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.84	1.82	1.77	1.72
2.61	2.51	2.41	2.30	2.23	2.18	2.12	2.08	2.05	1.98	1.91
3.13	2.99	2.85	2.70	2.60	2.54	2.45	2.40	2.36	2.27	2.18
4.56	4.31	4.06	3.79	3.63	3.52	3.37	3.28	3.22	3.06	2.91
1.86	1.81	1.76	1.71	1.67	1.65	1.61	1.59	1.58	1.54	1.51
2.22	2.15	2.07	1.99	1.94	1.90	1.85	1.82	1.80	1.75	1.70
2.59	2.49	2.39	2.28	2.21	2.16	2.09	2.05	2.03	1.95	1.89
3.09	2.96	2.81	2.66	2.57	2.50	2.42	2.36	2.33	2.23	2.14
4.48	4.24	3.99	3.72	3.56	3.44	3.30	3.21	3.15	2.99	2.84
1.85	1.80	1.75	1.70	1.66	1.64	1.60	1.58	1.57	1.53	1.50
2.20	2.13	2.06	1.97	1.92	1.88	1.84	1.81	1.79	1.73	1.68
2.57	2.47	2.36	2.25	2.18	2.13	2.07	2.03	2.00	1.93	1.86
3.06	2.93	2.78	2.63	2.54	2.47	2.38	2.33	2.29	2.20	2.11
4.41	4.17	3.92	3.66	3.49	3.38	3.23	3.14	3.08	2.92	2.78

(Continued)

T-18 Tables

TABLE E*F* critical values (continued)

		Degrees of freedom in the numerator								
<i>p</i>		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	28	.100	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.87
		.050	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.24
		.025	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.61
		.010	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.12
		.001	13.50	8.93	7.19	6.25	5.66	5.24	4.93	4.50
	29	.100	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.86
		.050	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.22
		.025	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.59
		.010	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.09
		.001	13.39	8.85	7.12	6.19	5.59	5.18	4.87	4.45
	30	.100	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.85
		.050	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.21
		.025	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.57
		.010	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.07
		.001	13.29	8.77	7.05	6.12	5.53	5.12	4.82	4.39
	40	.100	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.79
		.050	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.12
		.025	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.45
		.010	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.89
		.001	12.61	8.25	6.59	5.70	5.13	4.73	4.44	4.02
	50	.100	2.81	2.41	2.20	2.06	1.97	1.90	1.84	1.76
		.050	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.07
		.025	5.34	3.97	3.39	3.05	2.83	2.67	2.55	2.46
		.010	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89
		.001	12.22	7.96	6.34	5.46	4.90	4.51	4.22	3.82
	60	.100	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.74
		.050	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.04
		.025	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41
		.010	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82
		.001	11.97	7.77	6.17	5.31	4.76	4.37	4.09	3.69
	100	.100	2.76	2.36	2.14	2.00	1.91	1.83	1.78	1.69
		.050	3.94	3.09	2.70	2.46	2.31	2.19	2.10	1.97
		.025	5.18	3.83	3.25	2.92	2.70	2.54	2.42	2.32
		.010	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69
		.001	11.50	7.41	5.86	5.02	4.48	4.11	3.83	3.44
	200	.100	2.73	2.33	2.11	1.97	1.88	1.80	1.75	1.66
		.050	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.93
		.025	5.10	3.76	3.18	2.85	2.63	2.47	2.35	2.26
		.010	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60
		.001	11.15	7.15	5.63	4.81	4.29	3.92	3.65	3.26
	1000	.100	2.71	2.31	2.09	1.95	1.85	1.78	1.72	1.64
		.050	3.85	3.00	2.61	2.38	2.22	2.11	2.02	1.95
		.025	5.04	3.70	3.13	2.80	2.58	2.42	2.30	2.20
		.010	6.66	4.63	3.80	3.34	3.04	2.82	2.66	2.53
		.001	10.89	6.96	5.46	4.65	4.14	3.78	3.51	3.13

TABLE E*F* critical values (continued)

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
1.84	1.79	1.74	1.69	1.65	1.63	1.59	1.57	1.56	1.52	1.48
2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.79	1.77	1.71	1.66
2.55	2.45	2.34	2.23	2.16	2.11	2.05	2.01	1.98	1.91	1.84
3.03	2.90	2.75	2.60	2.51	2.44	2.35	2.30	2.26	2.17	2.08
4.35	4.11	3.86	3.60	3.43	3.32	3.18	3.09	3.02	2.86	2.72
1.83	1.78	1.73	1.68	1.64	1.62	1.58	1.56	1.55	1.51	1.47
2.18	2.10	2.03	1.94	1.89	1.85	1.81	1.77	1.75	1.70	1.65
2.53	2.43	2.32	2.21	2.14	2.09	2.03	1.99	1.96	1.89	1.82
3.00	2.87	2.73	2.57	2.48	2.41	2.33	2.27	2.23	2.14	2.05
4.29	4.05	3.80	3.54	3.38	3.27	3.12	3.03	2.97	2.81	2.66
1.82	1.77	1.72	1.67	1.63	1.61	1.57	1.55	1.54	1.50	1.46
2.16	2.09	2.01	1.93	1.88	1.84	1.79	1.76	1.74	1.68	1.63
2.51	2.41	2.31	2.20	2.12	2.07	2.01	1.97	1.94	1.87	1.80
2.98	2.84	2.70	2.55	2.45	2.39	2.30	2.25	2.21	2.11	2.02
4.24	4.00	3.75	3.49	3.33	3.22	3.07	2.98	2.92	2.76	2.61
1.76	1.71	1.66	1.61	1.57	1.54	1.51	1.48	1.47	1.42	1.38
2.08	2.00	1.92	1.84	1.78	1.74	1.69	1.66	1.64	1.58	1.52
2.39	2.29	2.18	2.07	1.99	1.94	1.88	1.83	1.80	1.72	1.65
2.80	2.66	2.52	2.37	2.27	2.20	2.11	2.06	2.02	1.92	1.82
3.87	3.64	3.40	3.14	2.98	2.87	2.73	2.64	2.57	2.41	2.25
1.73	1.68	1.63	1.57	1.53	1.50	1.46	1.44	1.42	1.38	1.33
2.03	1.95	1.87	1.78	1.73	1.69	1.63	1.60	1.58	1.51	1.45
2.32	2.22	2.11	1.99	1.92	1.87	1.80	1.75	1.72	1.64	1.56
2.70	2.56	2.42	2.27	2.17	2.10	2.01	1.95	1.91	1.80	1.70
3.67	3.44	3.20	2.95	2.79	2.68	2.53	2.44	2.38	2.21	2.05
1.71	1.66	1.60	1.54	1.50	1.48	1.44	1.41	1.40	1.35	1.30
1.99	1.92	1.84	1.75	1.69	1.65	1.59	1.56	1.53	1.47	1.40
2.27	2.17	2.06	1.94	1.87	1.82	1.74	1.70	1.67	1.58	1.49
2.63	2.50	2.35	2.20	2.10	2.03	1.94	1.88	1.84	1.73	1.62
3.54	3.32	3.08	2.83	2.67	2.55	2.41	2.32	2.25	2.08	1.92
1.66	1.61	1.56	1.49	1.45	1.42	1.38	1.35	1.34	1.28	1.22
1.93	1.85	1.77	1.68	1.62	1.57	1.52	1.48	1.45	1.38	1.30
2.18	2.08	1.97	1.85	1.77	1.71	1.64	1.59	1.56	1.46	1.36
2.50	2.37	2.22	2.07	1.97	1.89	1.80	1.74	1.69	1.57	1.45
3.30	3.07	2.84	2.59	2.43	2.32	2.17	2.08	2.01	1.83	1.64
1.63	1.58	1.52	1.46	1.41	1.38	1.34	1.31	1.29	1.23	1.16
1.88	1.80	1.72	1.62	1.56	1.52	1.46	1.41	1.39	1.30	1.21
2.11	2.01	1.90	1.78	1.70	1.64	1.56	1.51	1.47	1.37	1.25
2.41	2.27	2.13	1.97	1.87	1.79	1.69	1.63	1.58	1.45	1.30
3.12	2.90	2.67	2.42	2.26	2.15	2.00	1.90	1.83	1.64	1.43
1.61	1.55	1.49	1.43	1.38	1.35	1.30	1.27	1.25	1.18	1.08
1.84	1.76	1.68	1.58	1.52	1.47	1.41	1.36	1.33	1.24	1.11
2.06	1.96	1.85	1.72	1.64	1.58	1.50	1.45	1.41	1.29	1.13
2.34	2.20	2.06	1.90	1.79	1.72	1.61	1.54	1.50	1.35	1.16
2.99	2.77	2.54	2.30	2.14	2.02	1.87	1.77	1.69	1.49	1.22